It is recommended that you spend about 30 minutes on this question.

2 A fine wire mesh has individual wires that are spaced very close together. See Fig. 2.1.

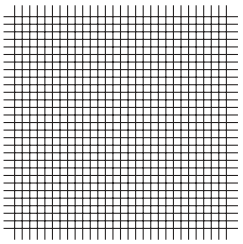

Fig. 2.1

The mesh behaves like two diffraction gratings placed at right angles to each other. The diffraction grating formula is $d \sin\theta = n\lambda$.

The spacing between the wires of the mesh is to be found accurately. Design a laboratory experiment using light of a single wavelength to determine the spacing between the wires. You may assume that the wavelength of the light is known.

You should draw a detailed labelled diagram showing the arrangement of your apparatus. In your account you should pay particular attention to

(a) the type of light source to be used, giving a reason for your choice,

(b) the procedure to be followed and the measurements that would be taken,

(c) how the measurements would be used to find values of θ,

(d) how the spacing between the wires would be deduced,

(e) any safety precautions you may take.

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